

PHD_THESIS — Systematic Reorganisation Plan

Detailed folder structure, migration map, and execution steps

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Abstract

The current `/root/phd_thesis` root contains **179 items** (121 files + 58 directories) at the top level, including 56 stray LaTeX auxiliary files (`.aux/.log/.out/.toc`), 14 slide study-note files scattered across 14 directories, and 20+ directories with overlapping or ambiguous purposes. This plan defines a clean 7-folder top-level taxonomy, specifies exactly where every existing item moves, and gives the exact shell commands to execute the migration safely — including git operations to track the 29 currently untracked technical reports.

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1. Diagnosis: What Is Wrong Now

1.1 Root-Level Clutter

121 files live at root — including 56 LaTeX build artefacts (`.aux`, `.log`, `.out`, `.toc`), 14 slide study-note PDFs/TEX/AUX sets, and 5 presentation PDFs that belong in subdirectories.

Table 1: Root-level clutter count

Type	Count	Problem
LaTeX build artefacts (<code>.aux/.log/.out/.toc/.nlo/.snm/.nav</code>)	56	Should go in build/
slideNN_study_note.* (tex+pdf+aux+log+out)	70	Should live in presentations/
slideNN_figures/ directories	10	Should live in presentations/
thesis.tex / synopsis.tex (orphan drafts)	2	Move to thesis_drafts/
*.mplstyle files	2	Move to scripts/styles/
tikz_master_styles.tex	1	Move to shared/styles/
reference.bib (global orphan)	1	Move to shared/
lock_pdf.py	1	Move to scripts/utilities/
*.qmd / _quarto.yml / _site/	6	Archive or docs/

1.2 Duplicate / Overlapping Directories

Table 2: Directories with overlapping purpose

Directories	Overlap
<code>new_thesis/</code> vs <code>new_thesis_edit/</code>	Two identical 13-chapter drafts; differ only by minor edits
<code>thesis_submission/</code> vs <code>new_thesis/</code> vs <code>2-mainmatter/</code>	Three independent thesis draft trees
<code>Z_Final_Thesis_data/</code> vs <code>data/</code> vs <code>simulations/</code>	Simulation data spread across three locations
<code>Z_Synthetic_Family_Library/</code> vs <code>Z_Synthetic_Family_Library_v2/</code>	Two versions, should be <code>data/synthetic_library/v1</code> and <code>v2</code>
<code>models/</code> vs <code>core_engines/</code>	Both contain ML <code>.pkl</code> model files
<code>ch4/</code> vs <code>ch6/</code> vs <code>Z_Final_Thesis_scripts/</code>	Chapter scripts in three places
<code>analytics_validation/</code> vs <code>data/validation/</code>	Validation results duplicated
<code>sambp_overleaf/</code> vs <code>sambp/sambp_diagrams/</code>	TikZ diagrams in two places
<code>docs/</code> vs <code>reports/</code> vs <code>PDF_COLLECTION/</code>	Reference PDFs in three locations
<code>presentation_delivery/</code> (empty) vs <code>presentations/</code>	One is empty

1.3 Git Tracking Gaps

29 technical reports are untracked (TR-16 to TR-37, TR-46 to TR-55). These 29 directories exist on disk with compiled PDFs but have never been committed to git. If the disk is lost or reset, all this work is gone.

1.4 SAMBP Report Naming

The **sambp/** directory has 59 report folders named **report3**, **report4**...**report62** with gaps at 59–61 and 63–65. There is no quick way to know which report covers which topic without opening each file. Folder names should encode the topic.

2. Proposed Top-Level Structure

The new structure uses **7 top-level folders** plus 3 root-level files. Every item currently in the root has exactly one destination.

```
phd_thesis/
|-- 01_thesis/           ← All thesis draft LaTeX (submission-ready)
|-- 02_papers/           ← All IEEE journal papers (6 papers)
|-- 03_technical_reports/ ← All 59 SAMBP TRs, grouped by phase
|-- 04_code/             ← All Python modules and scripts
|-- 05_data/             ← All simulation data, CSV, synthetic library
|-- 06_presentations/    ← Slides, TSA meeting, study notes
|-- 07_archive/          ← Superseded drafts, old Overleaf versions
|-- CLAUDE.md            ← Project instructions (keep at root)
|-- README.md            ← Top-level overview (keep at root)
'-- THESIS_CHAPTER_MAP.tex ← Master TR→chapter map (keep at root)
```

3. Detailed Target Structure — Folder by Folder

3.1 01_thesis/

Purpose: The single source of truth for the thesis LaTeX. One active draft; older versions go to 07_archive/.

```
01_thesis/
|-- active/              ← Current IIT Madras submission draft
|   |-- thesis_main.tex  ← Master file (from thesis_submission/)
|   |-- iitmdissertation.cls
|   |-- iitmdissertation.sty
|   |-- iitm.bst
|   |-- chapters/        ← ch01_introduction.tex ... ch13_conclusions.tex
|   |-- frontmatter/     ← abstract, acknowledgements, glossary
|   |-- appendices/
|   |-- figures/          ← All thesis figures
|   |-- images/           ← iitm-logo.png etc.
|   |-- references/
|   |   '-- thesis.bib
|   '-- styles/
|       |-- tikz_master_styles.tex
|       |-- thesis.mplstyle
|       '-- thesis_v2.mplstyle
|-- synopsis/
|   '-- synopsis.tex
|-- chapter_map/
|   '-- THESIS_CHAPTER_MAP.tex
'-- build/                ← LaTeX build artefacts (gitignored)
    '-- .gitkeep
```

Sources:

- thesis_submission/ → 01_thesis/active/
- synopsis.tex → 01_thesis/synopsis/

- THESIS_CHAPTER_MAP.tex → 01_thesis/chapter_map/
- tikz_master_styles.tex, *.mplstyle → 01_thesis/active/styles/
- Root thesis.tex, thesis*.log → 07_archive/root_orphans/

3.2 02_papers/

Purpose: All IEEE-format journal papers, one subfolder per paper. Each subfolder is self-contained (tex + bib + figures + Makefile).

```
02_papers/
|-- paper_a_syncoc_oc/          ← Paper A: Adaptive OC (IEEE TPWRD)
|   |-- main_paper.tex
|   |-- references.bib
|   |-- figures/
|   '-- Makefile
|-- paper_b_cascade_oc/        ← Paper B: Hilbert-Fortescue Cascade (IEEE TIE)
|   |-- main_paper_b.tex
|   |-- references.bib
|   |-- figures/
|   '-- Makefile
|-- paper_c_unified_sambp/     ← Paper C: Unified SAMBP (IEEE TPWRS)
|   |-- main_paper_c.tex
|   |-- references_c.bib
|   |-- figures/
|   '-- Makefile
|-- paper_d_6th_order/         ← 6th-order Park dq validation (IEEE)
|   ← from ieee_6th_order/
|-- paper_e_hif_tpwr/          ← HIF Impedance-Aware OC (IEEE TPWRD)
|   ← from ieee_hif_tpwr/ + sambp/hif_paper/
|-- paper_f_seq_estim/         ← Sequence Component Estimation (IEEE)
|   ← from ieee_seq_estim/
|-- paper_g_coord_gen/         ← Coordinated Adaptive OC (IEEE)
|   ← from ieee_coord_gen/
'-- paper_h_line_diff/         ← IBR-Adaptive 87L Line Diff (IEEE TIE)
    ← from ieee_line_diff/
```

Naming convention: paper_{letter}_{short_topic}/ Letters A–C are the main journal sequence; D–H are supporting papers.

3.3 03_technical_reports/

Purpose: All 59 SAMBP Technical Reports, grouped into 7 phase subfolders. Each TR folder is renamed to encode the topic.

```
03_technical_reports/
|-- phase_1_sg_framework/      ← TR-03 to TR-11 (SG core + MC)
|   |-- TR03_87T_87L_foundation/
|   |-- TR04_87B_bus_differential/
|   |-- TR05_system_integration_v1/
|   |-- TR06_iec61850_goose/
|   |-- TR07_double_busbar/
|   |-- TR08_monte_carlo_4k/
|   |-- TR09_stage2_recalibration/
|   |-- TR10_ct_saturation/
|   '-- TR11_hif_sensitivity/
```

```

|-- phase_2_87L_deep/          ← TR-12 to TR-27
|   |-- TR12_87L_magnitude_override/
|   |-- TR13_87L_pi_section/
|   |-- TR14_busbar_transfer_goose/
|   |-- TR15_87L_voltage_pi_section/
|   |-- TR16_87B_87L_coordination/
|   |-- TR17_87L_adaptive_threshold/
|   |-- TR18_mho_distance_21/
|   |-- TR19_simultaneous_transfer/
|   |-- TR20_frequency_adaptive_87L/
|   |-- TR21_harmonic_discrimination/
|   |-- TR22_87LN_negative_sequence/
|   |-- TR23_TDCS_87L/
|   |-- TR24_adaptive_TDCS/
|   |-- TR25_87LN0_zero_sequence/
|   |-- TR26_IBR_monte_carlo_15k/
|   '-- TR27_IBR_87L_integration/
|-- phase_3_distance/          ← TR-28 to TR-37
|   |-- TR28_synthesis_TR03_TR27/
|   |-- TR29_IBR_impedance_trajectory/
|   |-- TR30_cross_polarised_mho/
|   |-- TR31_zero_seq_compensation/
|   |-- TR32_directional_elements_67/
|   |-- TR33_POTT_IBR_lines/
|   |-- TR34_quadrilateral_distance/
|   |-- TR35_IBR_distance_coordination/
|   |-- TR36_auto_reclose_IBR/
|   '-- TR37_distance_MC_reliability/
|-- phase_4_microgrid/         ← TR-38 to TR-42
|   |-- TR38_GFL_GFM_characterisation/
|   |-- TR39_islanding_detection/
|   |-- TR40_adaptive_OC_microgrid/
|   |-- TR41_87B_microgrid/
|   '-- TR42_microgrid_integration/
|-- phase_5_digital_twin/      ← TR-43 to TR-45
|   |-- TR43_DT_architecture/
|   |-- TR44_EKF_estimation/
|   '-- TR45_DT_validation/
|-- phase_6_ML_advanced/       ← TR-46 to TR-55
|   |-- TR46_PINN_fault_classification/
|   |-- TR47_feature_engineering/
|   |-- TR48_online_learning/
|   |-- TR49_ML_integration/
|   |-- TR50_WAPC/
|   |-- TR51_settings_optimisation/
|   |-- TR52_cybersecurity_GOOSE/
|   |-- TR53_HIL_protocol/
|   |-- TR54_deployment_commissioning/
|   '-- TR55_thesis_synthesis/
'-- phase_7_IBR_extension/     ← TR-56, 57, 58, 62 (+ future 59-65)
    |-- TR56_DFIG_piecewise_ODE/
    |-- TR57_SPRT_87T_IBR/
    |-- TR58_EAC_CUEP_relay78/

```

```

|-- TR62_PV_2param_87L/
|-- TR59_DFIG_virtual_87G/      ← placeholder (planned)
|-- TR60_relay40_LOE_GFM/      ← placeholder
|-- TR61_64G_stator_earth/     ← placeholder
|-- TR63_relay81_IFDI/         ← placeholder
|-- TR64_87T_IBR_fullMC/       ← placeholder
'-- TR65_generator_suite/      ← placeholder

```

Internal structure of each TR folder (unchanged):

```

TR56_DFIG_piecewise_ODE/
|-- main_report56.tex
|-- references56.bib
|-- Makefile
|-- figures/
'-- main_report56.pdf

```

3.4 04_code/

Purpose: All Python source code, organised by protection function then by layer (models, estimation, detection, evaluation, adaptation). No ML .pkl model files here — those go to 05_data/models/.

```

04_code/
|-- sambp/                                ← Core SAMBP protection modules
|   |-- sync_oc/                          ← from sambp/sambp_sync_oc/
|   |   |-- models/
|   |   |-- inverse_estimation/
|   |   |-- evaluation/
|   |   |-- adaptation/
|   |   '--- signal_processing/
|   |-- line_diff/                        ← from sambp/sambp_line_diff/
|   |-- transformer_diff/                 ← from sambp/sambp_transformer_diff/
|   |-- bus_diff/                         ← from sambp/sambp_bus_diff/
|   |-- generator/                       ← from sambp/sambp_generator/
|   |-- distance/                        ← from sambp/sambp_distance/
|   |-- coordination/                    ← from sambp/sambp_coordination/
|   |-- double_bus/                      ← from sambp/sambp_double_bus/
|   '--- system/                          ← from sambp/sambp_system/
|-- chapter_scripts/                     ← Chapter-specific simulation scripts
|   |-- ch4_IPR/                          ← from ch4/ + Z_Final_Thesis_scripts/ch4_IPR/
|   |-- ch5_microgrid/                    ← from Z_Final_Thesis_scripts/ch5_Microgrid/
|   '--- ch6_AI_ML/                       ← from ch6/ + Z_Final_Thesis_scripts/ch6_AI_ML/
|-- data_processing/                     ← from data_processing/ (generators, CSV tools)
|-- analytics/                           ← from analytics_validation/ (audit, validation)
|-- utilities/                           ← from utilities/ + scripts/
|   |-- preview_tikz.py
|   |-- organize_thesis.sh
|   |-- sync_thesis.sh
|   |-- lock_pdf.py
|   |-- generate_synopsis.py
|   '--- ...
|-- notebooks/                           ← from notebooks/
|   |-- C6_Validation_Sandbox.ipynb

```

```
|  '-- Inverse_Source_modelling.ipynb
'-- styles/
|  '-- thesis.mplstyle
'-- thesis_v2.mplstyle
```

3.5 05_data/

Purpose: All data artefacts — simulation outputs, CSV files, trained ML models, synthetic waveform libraries. Nothing here is source code.

```
05_data/
|-- simulation_results/          ← from data/simulations/ + Z_Final_Thesis_data/
|  |-- ch4_IPR/
|  |-- ch5_microgrid/
|  '-- ch6_AI_ML/
|-- fault_waveforms/           ← from data/chapter_4/ + data/processing/
|  |-- pscad_fault_data.csv
|  |-- hif_fault_data.csv
|  |-- sg_fault_data.csv
|  '-- fault_data.csv
|-- synthetic_library/          ← from Z_Synthetic_Family_Library*
|  |-- v1/
|  |  |-- waveforms/
|  |  |-- metadata/
|  |  '-- benchmark_results/
|  '-- v2/
|     |-- waveforms/
|     |-- metadata/
|     '-- benchmark_results/
|-- validation_results/         ← from data/validation/ + analytics_validation/
|  |-- Master_100_Sim_Results.csv
|  |-- batch_validation_results.csv
|  '-- ici_audit_results.csv
|-- trained_models/             ← from models/ + core_engines/ (*.pkl files)
|  |-- ibr_fault_classifier.pkl
|  |-- feature_scaler.pkl
|  |-- model_a_magnitude.pkl
|  |-- model_b_entropy.pkl
|  |-- scaler_a.pkl
|  '-- scaler_b.pkl
'-- figures_archive/            ← from FIGURE_ARCHIVE/ + Z_Final_Thesis_figures/
|  |-- rendered/                ← PNG/PDF rendered outputs
|  '-- sources/                 ← TikZ .tex figure sources
```

3.6 06_presentations/

Purpose: All slides, study notes, and TSA meeting materials.

```
06_presentations/
|-- tsa_meeting/                ← from root TSA_meeting_*.tex/pdf
|  |-- TSA_meeting_Eluvathingal.tex
|  |-- TSA_meeting_Eluvathingal.pdf
|  |-- TSA_notes_Eluvathingal.tex
|  '-- TSA_notes_Eluvathingal.pdf
```

```

|-- synopsis_slides/                ← from presentations/*.pptx
|   |-- Anoop_PhD_Synopsis.pptx
|   '-- Anoop_Detailed_PhD_Synopsis.pptx
|-- sambp_overview/
|   |-- sambp_presentation_notes.pdf
|   '-- sambp_speaker_notes.pdf
'-- study_notes/                    ← from root slide03_* ... slide16_*
    |-- slide03/
    |   |-- slide03_study_note.tex
    |   |-- slide03_study_note.pdf
    |   '-- figures/
    |-- slide04/ ... slide16/        ← same pattern for 04-16
    '-- figures/                    ← from slide07_figures ... slide16_figures

```

3.7 07_archive/

Purpose: Superseded drafts and experimental work that is NOT deleted but is NOT part of the active pipeline. Read-only; nothing here is compiled or run as part of the thesis workflow.

```

07_archive/
|-- thesis_drafts/
|   |-- new_thesis/                  ← 13-chapter extended draft
|   |-- new_thesis_edit/            ← edited version of above
|   |-- 2-mainmatter/               ← earliest chapter-by-chapter draft
|   |-- thesis.tex                  ← orphan root draft
|   '-- sambp_overleaf/             ← Overleaf full-report draft
|-- reports_old/
|   |-- sambp_overleaf/sambp_full_report/ ← older compiled SAMBP report
|   '-- reports/ (syncoc_deviation, research_gap)
|-- quarto_site/                    ← from _site/, audit.qmd, *.qmd, _quarto.yml
|-- old_pdfs/
|   |-- SAMBP_Research_Gap_Plan.pdf
|   |-- SAMBP_SyncOC_Deviation_Analysis.pdf
|   '-- PDF_COLLECTION/
|-- scratch/
|   |-- figures_orphan/              ← visualization/, images/, figures_thesis/
|   |-- build_artefacts/             ← build/
|   '-- simulations_old/             ← simulations/plots/
'-- root_orphans/                    ← stray tex/log at root that match nothing
    |-- texput.log
    '-- reference.bib

```

4. File-by-File Migration Map

Current Location	New Location	Action
— <i>Root-level files</i> —		
thesis_submission/	01_thesis/active/	Move
THESIS_CHAPTER_MAP.tex	01_thesis/chapter_map/	Move
synopsis.tex	01_thesis/synopsis/	Move
tikz_master_styles.tex	01_thesis/active/styles/	Move
thesis.mplstyle,	01_thesis/active/styles/	Move
thesis_v2.mplstyle		
thesis.tex	07_archive/thesis_drafts/	Move
reference.bib	07_archive/root_orphans/	Move
lock_pdf.py	04_code/utilities/	Move
README.md	(keep at root)	Stay
RESEARCH_GAP_PLAN.md	(keep at root)	Stay
texput.log, thesis*.log	Delete	Delete
THESIS_CHAPTER_MAP.aux/.log/.out/.toc	Delete	Delete
— <i>Slide study notes (03–16)</i> —		
slide03_study_note.*	06_presentations/study_notes/slide03/	Move
slide04 ... slide16	06_presentations/study_notes/slideNN/	Move
slide07_figures/	06_presentations/study_notes/slideNN/figures/	Move
...slide16_figures/		
— <i>TSA materials</i> —		
TSA_meeting_Eluvathingal.*	06_presentations/tsa_meeting/	Move
TSA_notes_Eluvathingal.*	06_presentations/tsa_meeting/	Move
presentations/*.pptx	06_presentations/synopsis_slides/	Move
sambp_presentation_notes.pdf	06_presentations/sambp_overview/	Move
sambp_speaker_notes.pdf	06_presentations/sambp_overview/	Move
— <i>IEEE papers</i> —		
sambp/sambp_sync_oc/paper/	02_papers/paper_a_syncoc_oc/	Move
sambp/sambp_sync_oc/paper_b/	02_papers/paper_b_cascade_oc/	Move
sambp/sambp_sync_oc/paper_c/	02_papers/paper_c_unified_sambp/	Move
ieee_6th_order/	02_papers/paper_d_6th_order/	Move
ieee_hif_tpwr/	02_papers/paper_e_hif_tpwr/	Move
sambp/hif_paper/	02_papers/paper_e_hif_tpwr/older_draft	Merge
ieee_seq_estim/	02_papers/paper_f_seq_estim/	Move
ieee_coord_gen/	02_papers/paper_g_coord_gen/	Move
ieee_line_diff/	02_papers/paper_h_line_diff/	Move
— <i>Technical reports</i> —		
sambp/report3/	03_technical_reports/phase_1_sg_framework/	Move+Rename
sambp/report4/	.../TR04_87B_bus_differential/	Move+Rename
sambp/report5/	.../TR05_system_integration_v1/	Move+Rename
sambp/report6/	.../TR06_iec61850_goose/	Move+Rename
sambp/report7/	.../TR07_double_busbar/	Move+Rename
sambp/report8/	.../TR08_monte_carlo_4k/	Move+Rename
sambp/report9/	.../TR09_stage2_recalibration/	Move+Rename
sambp/report10/	.../TR10_ct_saturation/	Move+Rename
sambp/report11/	.../TR11_hif_sensitivity/	Move+Rename
sambp/report12–15/	.../phase_2_87L_deep/TR12–15.../	Move+Rename
sambp/report16–27/	.../phase_2_87L_deep/TR16–27.../	Move+Rename

Current Location	New Location	Action
sambp/report28-37/	.../phase_3_distance/TR28-37.../	Move+Rename
sambp/report38-42/	.../phase_4_microgrid/TR38-42.../	Move+Rename
sambp/report43-45/	.../phase_5_digital_twin/TR43-45.../	Move+Rename
sambp/report46-55/	.../phase_6_ML_advanced/TR46-55.../	Move+Rename
sambp/report56,57,58,62/	.../phase_7_IBR_extension/TR56.../	Move+Rename
— Code modules —		
sambp/sambp_sync_oc/	04_code/sambp/sync_oc/	Move
sambp/sambp_line_diff/	04_code/sambp/line_diff/	Move
sambp/sambp_transformer_diff/	04_code/sambp/transformer_diff/	Move
sambp/sambp_bus_diff/	04_code/sambp/bus_diff/	Move
sambp/sambp_generator/	04_code/sambp/generator/	Move
sambp/sambp_distance/	04_code/sambp/distance/	Move
sambp/sambp_coordination/	04_code/sambp/coordination/	Move
sambp/sambp_double_bus/	04_code/sambp/double_bus/	Move
sambp/sambp_system/	04_code/sambp/system/	Move
ch4/, ch6/	04_code/chapter_scripts/	Merge
Z_Final_Thesis_scripts/	04_code/chapter_scripts/	Merge
analytics_validation/	04_code/analytics/	Move
data_processing/	04_code/data_processing/	Move
utilities/	04_code/utilities/	Merge
scripts/	04_code/utilities/	Merge
notebooks/	04_code/notebooks/	Move
core_engines/	Split: .pkl → 05_data/; .py → 04_code/	Split
models/	05_data/trained_models/	Move
— Data —		
data/simulations/	05_data/simulation_results/	Move
data/chapter_4/	05_data/fault_waveforms/	Move
data/processing/	05_data/fault_waveforms/	Merge
data/validation/	05_data/validation_results/	Move
Z_Final_Thesis_data/	05_data/simulation_results/	Merge
Z_Synthetic_Family_Library/	05_data/synthetic_library/v1/	Move
Z_Synthetic_Family_Library_v2/	05_data/synthetic_library/v2/	Move
FIGURE_ARCHIVE/	05_data/figures_archive/	Move
Z_Final_Thesis_figures/	05_data/figures_archive/chapter_sources	Merge
— Archive —		
new_thesis/	07_archive/thesis_drafts/new_thesis/	Move
new_thesis_edit/	07_archive/thesis_drafts/new_thesis_edit/	Move
2-mainmatter/	07_archive/thesis_drafts/2-mainmatter/	Move
sambp_overleaf/	07_archive/thesis_drafts/sambp_overleaf/	Move
docs/	07_archive/reference_pdfs/	Move
reports/	07_archive/reports_old/	Move
PDF_COLLECTION/	07_archive/old_pdfs/	Move
_site/, *.qmd, _quarto.yml	07_archive/quarto_site/	Move
backups/	07_archive/backups/	Move
build/	07_archive/build_artefacts/	Move
simulations/	07_archive/scratch/	Move
visualization/	07_archive/scratch/	Move
presentation_delivery/	Delete	Delete
(empty)		

5. Git Strategy

5.1 Problem: 29 Untracked Reports Must Be Committed First

CRITICAL: Git must be updated BEFORE any files are moved. If you move files and then commit, git sees “deleted old + added new” and loses history. The correct sequence is:

Step 1: Commit the 29 untracked reports.

Step 2: Use `git mv` to move everything (history preserved).

Step 3: Commit the reorganisation.

5.2 Step 1 — Commit Untracked Reports

Run once to stage all 29 untracked TR folders:

```
cd /root/phd_thesis
```

```
# Add all untracked reports (TR-16 to TR-37, TR-46 to TR-55)
for n in 16 17 18 19 20 21 22 23 24 25 26 27 \
        28 29 30 31 32 33 34 35 36 37 \
        46 47 48 49 50 51 52 53 54 55; do
    git add sambp/report${n}/
done
```

```
git commit -m "Add untracked TR-16 to TR-55 (29 reports, 87L/distance/microgrid/DT/ML phases)"
```

5.3 Step 2 — Create New Directory Structure

```
mkdir -p 01_thesis/{active/{chapters,frontmatter,appendices,figures,images,
    references,styles},synopsis,chapter_map,build}
mkdir -p 02_papers/paper_{a_syncoc_oc,b_cascade_oc,c_unified_sambp,
    d_6th_order,e_hif_tpwr,d_f_seq_estim,g_coord_gen,h_line_diff}
mkdir -p 03_technical_reports/{phase_1_sg_framework,phase_2_87L_deep,
    phase_3_distance,phase_4_microgrid,phase_5_digital_twin,
    phase_6_ML_advanced,phase_7_IBR_extension}
mkdir -p 04_code/{sambp/{sync_oc,line_diff,transformer_diff,bus_diff,
    generator,distance,coordination,double_bus,system},
    chapter_scripts/{ch4_IPR,ch5_microgrid,ch6_AI_ML},
    data_processing,analytics,utilities,notebooks,styles}
mkdir -p 05_data/{simulation_results/{ch4_IPR,ch5_microgrid,ch6_AI_ML},
    fault_waveforms,synthetic_library/{v1,v2},
    validation_results,trained_models,figures_archive/{rendered,sources}}
mkdir -p 06_presentations/{tsa_meeting,synopsis_slides,sambp_overview,
    study_notes}
mkdir -p 07_archive/{thesis_drafts,reports_old,quarto_site,
    old_pdfs,scratch,backups,build_artefacts,root_orphans}
```

5.4 Step 3 — Move Technical Reports (with rename)

```
BASE=03_technical_reports
```

```
# Phase 1 - SG Framework
```

```
declare -A TR_NAMES=(
```

```
    [3]="TR03_87T_87L_foundation"
```

```
    [4]="TR04_87B_bus_differential"
```

```

[5]="TR05_system_integration_v1"      [6]="TR06_iec61850_goose"
[7]="TR07_double_busbar"              [8]="TR08_monte_carlo_4k"
[9]="TR09_stage2_recibration"         [10]="TR10_ct_saturation"
[11]="TR11_hif_sensitivity"
)
PHASE=phase_1_sg_framework
for n in 3 4 5 6 7 8 9 10 11; do
    git mv sambp/report${n} ${BASE}/${PHASE}/${TR_NAMES[$n]}
done

# Phase 2 - 87L Deep
declare -A TR_NAMES2=(
    [12]="TR12_87L_magnitude_override" [13]="TR13_87L_pi_section"
    [14]="TR14_busbar_transfer_goose"  [15]="TR15_87L_voltage_pi_section"
    [16]="TR16_87B_87L_coordination"    [17]="TR17_87L_adaptive_threshold"
    [18]="TR18_mho_distance_21"         [19]="TR19_simultaneous_transfer"
    [20]="TR20_frequency_adaptive_87L"  [21]="TR21_harmonic_discrimination"
    [22]="TR22_87LN_negative_sequence"  [23]="TR23_TDCS_87L"
    [24]="TR24_adaptive_TDCS"           [25]="TR25_87LN0_zero_sequence"
    [26]="TR26_IBR_monte_carlo_15k"     [27]="TR27_IBR_87L_integration"
)
PHASE=phase_2_87L_deep
for n in 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27; do
    git mv sambp/report${n} ${BASE}/${PHASE}/${TR_NAMES2[$n]}
done

# (same pattern for phases 3-7 - see full script in 04_code/utilities/)

```

5.5 Step 4 — Move Code Modules

```

# SAMBP Python modules
git mv sambp/sambp_sync_oc           04_code/sambp/sync_oc
git mv sambp/sambp_line_diff         04_code/sambp/line_diff
git mv sambp/sambp_transformer_diff  04_code/sambp/transformer_diff
git mv sambp/sambp_bus_diff          04_code/sambp/bus_diff
git mv sambp/sambp_generator         04_code/sambp/generator
git mv sambp/sambp_distance          04_code/sambp/distance
git mv sambp/sambp_coordination      04_code/sambp/coordination
git mv sambp/sambp_double_bus        04_code/sambp/double_bus
git mv sambp/sambp_system            04_code/sambp/system

```

5.6 Step 5 — Move Papers

```

git mv ieee_6th_order               02_papers/paper_d_6th_order
git mv ieee_hif_tpwd                02_papers/paper_e_hif_tpwd
git mv ieee_seq_estim               02_papers/paper_f_seq_estim
git mv ieee_coord_gen               02_papers/paper_g_coord_gen
git mv ieee_line_diff               02_papers/paper_h_line_diff
git mv sambp/hif_paper              02_papers/paper_e_hif_tpwd/older_draft

```

5.7 Step 6 — Move Thesis and Presentations

```

git mv thesis_submission             01_thesis/active
git mv THESIS_CHAPTER_MAP.tex       01_thesis/chapter_map/

```

```

git mv synopsis.tex                01_thesis/synopsis/
git mv tikz_master_styles.tex      01_thesis/active/styles/

# Slide study notes (03-16)
for n in $(seq -w 3 16); do
  mkdir -p 06_presentations/study_notes/slide${n}
  git mv slide${n}_study_note.tex  06_presentations/study_notes/slide${n}/
  git mv slide${n}_study_note.pdf  06_presentations/study_notes/slide${n}/ 2>/dev/null
  for ext in aux log out toc; do
    rm -f slide${n}_study_note.${ext}
  done
done
for n in 07 08 10 11 12 13 14 15 16; do
  git mv slide${n}_figures 06_presentations/study_notes/slide${n}/figures
done

# TSA materials
for ext in tex pdf; do
  git mv TSA_meeting_Eluvathingal.${ext} 06_presentations/tsa_meeting/
  git mv TSA_notes_Eluvathingal.${ext}    06_presentations/tsa_meeting/
done

```

5.8 Step 7 — Archive Old Drafts

```

git mv new_thesis          07_archive/thesis_drafts/
git mv new_thesis_edit     07_archive/thesis_drafts/
git mv 2-mainmatter        07_archive/thesis_drafts/
git mv sambp_overleaf      07_archive/thesis_drafts/
git mv docs                07_archive/reference_pdfs
git mv reports             07_archive/reports_old
git mv PDF_COLLECTION      07_archive/old_pdfs
git mv backups             07_archive/
git mv build               07_archive/build_artefacts
git mv simulations         07_archive/scratch/simulations_old
git mv visualization       07_archive/scratch/

```

5.9 Step 8 — Delete Clutter

```

# Delete all LaTeX build artefacts at root
rm -f THESIS_CHAPTER_MAP.{aux,log,out,toc}
rm -f TSA_meeting_Eluvathingal.{aux,log,nav,out,snm,toc}
rm -f TSA_notes_Eluvathingal.{aux,log,out,toc}
rm -f texput.log thesis.log thesis_main.log thesis.nlo
rm -f SAMBP_Research_Gap_Plan.pdf SAMBP_SyncOC_Deviation_Analysis.pdf

# Delete empty directory
rm -rf presentation_delivery
rm -rf __pycache__ _assets _site .quarto .ipynb_checkpoints

```

5.10 Step 9 — Update .gitignore

Add to .gitignore:

```
# LaTeX build artefacts
```

```
**/*.aux
**/*.log
**/*.out
**/*.toc
**/*.blg
**/*.bbl
**/*.nav
**/*.snm
**/*.nlo
**/*.ist
**/*.acn
**/*.glo
**/*.lof
**/*.lot
01_thesis/build/
03_technical_reports/**/*.figures/*.pdf # auto-generated figures

# Python
**/__pycache__/
**/*.pyc
**/.ipynb_checkpoints/

# Virtual environments
phd_env/
venv/
.venv/
```

5.11 Step 10 — Final Commit

```
git add -A
git commit -m "Reorganise PHD_THESIS into 7-folder taxonomy (01_thesis through 07_archive)"
```

6. Before / After Comparison

Table 4: Root-level directory count: before vs. after

Metric	Before	After
Top-level directories	58	7
Top-level files	121	3
Total root-level items	179	10
Untracked TR folders	29	0
Stray .aux/.log files at root	56	0
Slide study-note sets at root	14	0
Overlapping thesis draft trees	3	1 active + archive
TR folders with opaque names	59	0 (all topic-named)

7. Rules Going Forward

R1. New TR: Create in `03_technical_reports/phase_{N}/TR{XX}_{topic}/`. Commit immediately after writing.

R2. New code: Goes in `04_code/samp/{function}/`. Never at root.

R3. Build artefacts: `.aux/.log/.out/.toc` go into `01_thesis/build/` or `03_technical_reports/.../build/` — all gitignored.

R4. New paper: Create in `02_papers/paper_{letter}_{topic}/`.

R5. New data file: Goes in `05_data/` under the appropriate subcategory. Never committed alongside code.

R6. Presentations: All slides go into `06_presentations/`.

R7. Nothing at root except `README.md`, `RESEARCH_GAP_PLAN.md`, `.gitignore`, and `CLAUDE.md`.